

Consultation room 2030

Continuity of care from hospital to home

1. General information

1.a Applicants

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/Applicant]
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Erasmus MC						
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Planned start date of seed funding: 01/09/2022

Primary Convergence theme: *Technology Supported Transitions in Health(care)*

Secondary Convergence theme (if applicable): *Improving Health Journeys*

Tertiary Convergence theme (if applicable): Choose an item.

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: *Digital health technologies; consultation room; co-design; embedding; formative evaluation; reimbursement structures; data-infrastructure*

1.b Summary (277 of max 300 words)

Sustainability of healthcare is under pressure because of staff shortages, financial limitations, and societal expectations. Technological innovations could offer solutions, but many innovations in healthcare are characterized by non-adoption or by failed attempts to scale up, spread distantly, or sustain the innovation long term at the organization or system level. Embracing the possibilities of digital health technologies requires rethinking of the processes in the traditional consultation room. This Flagship takes up the challenge of designing the Consultation Room of the future in- and outside the hospital through building on existing initiatives (level 1), connecting, comparing, and integrating what is required for these initiatives to become embedded sustainably in the broader healthcare system (level 2) and by extending our network while facilitating learning and identifying the most promising solutions (level 3). Within this Flagship 7 PhDs, a Post-Doc and researchers from Erasmus Medical Center, Erasmus University Rotterdam and TU Delft contribute to the aim to shape the (conditions for the) continuity of care from hospital to home through digital technologies, in which: 1) digital technologies for various conditions are integrated via co-design in care pathways; 2) the embedding of these technologies is addressed from an interdisciplinary perspective; (3) the impact of the digital technologies is continuously monitored through formative evaluation developed for different types of outcome measures and soft impacts. One of the goals is to become a Centre of Expertise on digital health by identifying the most promising approaches to structurally embed such technologies in care networks and facilitate inter-organizational learning. This will facilitate a sustainable, interdisciplinary research group with the ability to provide solutions for current healthcare issues and to proactively facilitate future changes in healthcare.

1.c Public Summary (287 of max 300 words)

Healthcare is changing along with the wishes of patients to receive care at home, manage their own condition, being informed, and take decisions together with the healthcare provider. Also considering the challenges of staff shortages and financial limitations means that systemic changes of healthcare are necessary. These changes could be initiated and supported by digital technologies, in and outside the consultation room. Implementing digital health technologies in clinical practice is considered challenging and comes with organizational, social, and practical challenges. This requires rethinking the processes within the consultation room, as well as changes in the system in which healthcare functions. Researchers and clinicians from Erasmus Medical Center, Erasmus University Rotterdam and TU Delft contribute to the aim to shape the (conditions for the) continuity of care from hospital to home through digital technologies. This Flagship takes up the challenge of designing the Consultation Room of the future through building on existing initiatives (level 1), connecting, comparing, and integrating what is required for these initiatives to become embedded sustainably in the broader healthcare system (level 2) and by extending our network while facilitating learning and identifying the most promising solutions (level 3). With this Flagship we will have an impact on the clinical process with the embedding of technologies in specific contexts, designing future pathways and sharing lessons learned between healthcare domains. For the broader embedding we experiment with alternative payment methods for health technologies and evaluate the creation of a platform for the provision of care at a distance by connecting all aspects of the care process. The Flagship will continue as a centre of expertise for sustainable digital solutions sharing our knowledge, methodological toolbox, and regulatory/organizational understanding to increase the acceptability and use of digital technologies.

2. Relevance to Convergence Health & Technology mission 2498 of max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

The long-term sustainability of healthcare is under increasing pressure. Recent reports of the Netherlands Scientific Council for Government Policy [1] and the Social and Economic Council of the Netherlands [2] emphasize future healthcare system sustainability problems due to increasing staff shortages, financial limitations, and societal expectations. These issues become harder to address as healthcare costs and burden of illness keep increasing due to aging population and multimorbidity, while personnel shortages proliferate.

Digital health technologies can mitigate these substantive challenges. Defined broadly as a collection of technologies spanning genomics (genome reading and genome editing), digital medicine (ranging from telemedicine to VR), AI (from natural language processing to predictive analytics), and robotics [5], we focus in this Flagship on “digital products and services intended for use in the diagnosis, prevention, monitoring and treatment of a disease, condition or syndrome” (5: p.46). Digital health technologies can support, and in cases successfully replace, certain patient-physician/nurse interactions, leading to major health gains, improved efficiency of care processes, early identification of conditions, decreased workload and personalized health journeys [3, 4]. They can also empower patients to manage their own conditions within a controlled environment of remote care and monitoring.

The potential of digital technologies has also been embraced in the Erasmus MC as various promising initiatives have been started both inside and outside the consultation room. Successful examples in the consultation room are dashboards and immersive extended reality (XR) applications with predictive models and ePRO-results providing individualized outcome data supporting best decisions. Examples outside the consultation room include telemonitoring of pregnant women at risk for hypertensive complications, home monitoring of patients with idiopathic pulmonary fibrosis via real-time wireless home spirometry, digital preoperative screening tools improving differentiation between low-risk and high-risk patients to determine when preoperative anesthesia visits can be done at home, and remote oncology care enabling patients in the palliative phase to receive optimal care at home [7-9].

Digital technologies in the consultation room of the future

Embracing the possibilities of digital health technologies requires rethinking of the processes in the traditional consultation room. The consultation room (outpatient clinic) is a central place where contact between patient and healthcare providers is established and where shared decisions are made. Inside the consultation room a development has started with dashboards and immersive extended reality (XR) applications with predictive models and ePRO-results providing individualized outcome data empowering patients and supporting best decisions [37-40]. The Consultation Room of the future is, however, also rapidly extending outside of the traditional hospital walls. New (digital) technologies expand care processes and interactions with patients to new places. ***The future hospital is increasingly becoming a ‘hospital without boundaries’*** – that is, care becomes increasingly independent of location as patients receive blended types of care.

The development and further uptake of digital technologies and other technologies that enhance patient empowerment and interaction between patients and healthcare providers (e.g., XR-applications, prediction models, ePRO-structures) leads to a future Consultation Room that is both upgraded in the hospital and no longer confined to a single place. It functions as a 'boundary object' [32] that connects, mediates, and reconfigures care provision, at home and through technological extensions within the hospital. ***The Consultation Room of the future is about connecting different worlds and places:*** 1) the home environment of the patient and primary care with the hospital through home monitoring and teleconsultation); 2) different moments of care by developing prediction models for personalized counseling, early detection of adverse events or prevention of unnecessary hospital visits; 3) different IT- and data-infrastructures to enable information flow to/from patients homes to the hospital; 4) various modes of reimbursement to safeguard continuity between hospital care and technological support in the home environment.

Digital Health as systemic innovation

Implementing digital technologies in clinical practice is challenging [10-13] and comes with organizational, social, and practical hurdles that often interact in subtle ways [14-16]. The greatest barriers are not primarily technical in nature, but organizational and systemic [6, 37]. Therefore, a holistic research and implementation approach is needed, in which technology is studied ***in interaction with*** changing organizational structures, management processes, economic feasibility, societal impact, user perceptions, policy and governance [13,17]. The full uptake of digital technologies in healthcare should be viewed as ***a systemic innovation requiring new ways of providing, organizing, governing, and reimbursing care.*** Successful and sustainable use requires not only evidence of cost-effectiveness and health outcomes, but also expertise in co-design, management, governance, and sociology of innovation (e.g., changing roles of healthcare workers and patients).

In this Flagship, we lay the conceptual, methodological, and practical groundwork for the sustainable embedding of digital health technologies. We develop a multidisciplinary ecosystem involving key stakeholders, ranging from medical professionals, designers, technicians and patients to governance and regulatory bodies, health insurers, IT-support, and ethical and legal experts.

Taking leadership in the transition towards digital healthcare: a multi-level approach

Our ambition ***is to shape the Consultation Room of the future*** by taking leadership in the transition towards digital healthcare. This requires the ***further development of promising initiatives*** within the EMC, the ***connection and integration*** of these initiatives to identify more general lessons and best practices to generate sustainable organizational change, and ***expansion and scale-up*** to other hospitals in the regional network (details in 3a and 3b). We shape a future Consultation Room in which digital technologies are fully integrated in care provision to optimize patient support and empowerment, facilitate, and strengthen professional work, and contribute to continuity of care between hospital and home.

Contribution to the H&T Mission

The Flagship is closely aligned with the ambitions and aims of Convergence Health & Technology and the strategic, hospital-wide Erasmus MC program "Digitally Connected" (see 3c). Through building on existing initiatives (level 1), connecting, comparing and integrating what is required for these initiatives to become embedded sustainably in the broader healthcare system (level 2) and extending our network while facilitating learning and identifying the most promising solutions (level 3), the Flagship ***contributes to multiple aims: 1) enhancing prevention and early diagnostics to proactively maintain health; 2) develop integrated digital technologies allowing for earlier***

detection of medical risks; 3) enable more proactive monitoring at a distance. The Flagship also contributes to the Convergence aim of *life-long health and well-being* through its focus on patient and professional support through digital means.

2.b Synergy between disciplines

A Convergent Approach to Digital health as Systemic Innovation

Embedding digital health technologies sustainably is a key example of an interdisciplinary challenge requiring synergy between different disciplines and methodologies. ***This Flagship integrates medical and paramedical expertise*** on relevant health outcomes and targets, ***technical expertise*** on IT- and data-infrastructures, ***design expertise*** on co-design of technological innovations, ***sociological and organizational expertise*** on changing work processes and patient-provider interactions, ***health economic expertise*** on cost-effectiveness and value-based healthcare, ***governance and regulatory expertise*** on data ethics and legal requirements, and ***patient expertise*** to ensure digital innovations align with patients' needs. This Flagship is therefore a prime example of Convergence as it integrates experts, tools and methodologies from different disciplines, such as the medical and health sciences, technical and data sciences, and social sciences.

To enable a coherent overarching perspective, we integrate two theoretical frameworks that focus on ***holistic and integrated analysis*** of medical practices, technology, organizational requirements, data- and IT-infrastructures, and regulatory and governance requirements. ***This enables us to also anchor synergy on a theoretical level.*** The SELFIE-framework [19] distinguishes six [WHO] components on micro, meso, and macro levels. It provides a holistic conceptual framework to aid development, implementation, description, and evaluation of integrated care programs that are evidence-based, economically sustainable, integrated across domains and supported by appropriate payment schemes. We extend this framework with the NASSS-framework: an evidence-based, theory-informed, and pragmatic framework to help predict and evaluate the success of technology-supported health or social care programs [18, 20].

Institutionalizing Synergy within the Flagship

With this Flagship we develop synergy with existing innovations currently ongoing within the Erasmus MC. Connecting these hitherto fragmented projects provides input for necessary systemic changes for integration and embedding of digital health technologies. ***This Flagship builds on the strengths of existing innovation projects within the Erasmus MC in five domains of care: acute, chronic, complex, mental, and palliative care.*** Research leads of all projects are included in this Flagship consortium and offer in-cash contributions to the Flagship (see 3c and annex 1).

Simultaneously, we explicitly ***move beyond evaluations of individual projects.*** Through an interdisciplinary approach, we focus on studying and shaping the entire process and necessary conditions to embed digital health technologies sustainably in healthcare practices. We focus on the three different levels: learning from existing initiatives; identifying broader lessons and best practices through comparing and integrating; and extending the network to enable systemic national innovation (see 3a).

The different disciplinary perspectives are not only integrated in each work package and in each PhD trajectory (see 3c) but are also ***solidified through institutional structures and community building methods and tools*** (see 4d).

2.c Long term sustainability plan

Our Flagship ambition is *shaping the Consultation Room of the future by taking leadership in the transition towards digital healthcare*. To achieve this ambitious goal, the Flagship brings together multiple stakeholders and experts from different disciplines to establish a **vibrant ecosystem committed to taking the next step towards digital health**. Through targeted efforts in the five domains (acute, chronic, complex, mental, and palliative care) we develop a rich set of insights into the workings and effects of digital health technologies. We are inclusive in our approach and actively involve other ongoing and future initiatives in the EMC. We expect that the joint co-creation, embedding and formative evaluation of separate but interconnected projects through multidisciplinary teams of senior researchers and PhDs allows for broader and more integrated cross-domain analyses that can **act as a catalyst for new initiatives in digital health**. The experience, knowledge and research methods acquired during the flagship will be sustained as a **Center of Expertise on digital health in medicine** after the funding period of the Flagship. This Center of Expertise will be supported by the community built during the Flagship and serves as an innovation hub, learning environment and educational platform, equipping current and future students of all institutes with the skills to work in a transdisciplinary manner and drive forward the full potential of these technical advances. One of the PhDs is expected to grow into a tenure track as postdoc in year 5, thus safeguarding continuity. We will also actively incorporate other interested stakeholders, including policy makers, legislative experts and insurance companies to pave the way to structural embedding and reimbursement.

Risks involved and how to tackle them

Stakeholders could opt out due to lack of time

We expect this risk to be mitigated through the firm level of support expressed by all consortium partners. This support is also anchored through many in-cash and in-kind contributions (annex 1 and 2). The regulators involved have expressed their need for practice-based experience to develop and adjust their regulatory arrangements for digital health technologies. The involved representatives from the National Health Care Institute (ZiN) and the Health and Youth Care Inspectorate (IGJ) have formal academic positions at EUR; moreover, representatives from the Dutch Healthcare Authority (NZa), National Institute for Public Health and the Environment (RIVM) and health insurance confirmed their participation in the Reflective Board. Many private companies support the initiative (annex 2).

Research funding may not be granted

We expect the (inter)national funding landscape to provide ample opportunities for funding large-scale, interdisciplinary consortia of committed researchers dedicated to facilitating the sustainable uptake and embedding of digital health technologies. All institutions and consortium members are very successful in obtaining research grants and are widely acknowledged for their expertise (see 4a). We are convinced that we will profit from this strong basis and will obtain more prestigious grants, both among individual scholars and as a transdisciplinary consortium.

A renewed outbreak of the COVID pandemic

The COVID pandemic created immense pressure on health systems and researchers worldwide, but also accelerated technology use in medical care. A renewed outbreak could create challenges for remaining on track with the Flagship program. Given the expertise gained in using digital methods of communication ourselves, we are confident we will be able to continue the research in accordance with safety measures in case a renewed outbreak occurs (it could even be a catalyst for accelerated implementation of future consultation).

2.d Visibility, attracting talents and new partners

With this Flagship we create visibility in several ways. We already established a **website** (under construction) that will increase visibility of the Flagship. In **education** we are already using our collaborations (e.g., in 10-week bachelor course ESHPM on technology and innovation) for cases or research projects in education. The Flagship will function as an accelerator for joint educational activities. Concretely, ***we use the Consultation Room 2030 as a field lab in which students from TUD, EMC and EUR can do a research project or write their thesis.*** Students from different studies at EUR, EMC and TUD (such as healthcare management, medicine, and industrial design engineering) can apply for positions within this lab to write their theses. They will work on this in collaboration with the PhDs, postdoc and senior staff in the **Convergence Hubs** creating an environment in which we can meet and learn from each other (see 4d). A **joint symposium** will further establish the connections and learning environment and serves to create additional interest and visibility along partners and interested stakeholders.

External partners are engaged through the **Reflective Board** (see 4d) and our research projects. The Reflective Board consists of governance and regulatory actors, health insurers, members focused on data protection and ethics, and representatives from patient and professional associations, who meet twice a year to discuss developments and progress of the Flagship and provide feedback and recommendations to the Management Board and to specific projects. This will have a positive impact on our network and visibility. It can also lead to collaborations and funding from, for example, health care insurers or the government. Other external partners like manufacturers will have a role in the specific projects in which they collaborate. Research projects are encouraged to work with third parties to establish connections.

2.e Long term societal impact

Through studying digital health technologies in five medical domains simultaneously, as connecting and integrating insights and best practices, ***we expect that the societal impact of this Flagship will radiate in different directions.*** The projects are currently in TRL levels 5-7 and with our partners we want them to move towards TRL 9 (system launch). Our Flagship is expected to lead to professionals becoming more knowledgeable of (and attuned to) ways of providing care digitally when possible, patients becoming more empowered and trusted with the opportunities for better counseling, monitoring and prevention offered by digital technologies, regulatory and governance actors developing clearer frameworks for quality control of digital technologies. and health insurers developing innovative reimbursement schemes that incentivize digital health technologies. The Reflective Board and the discussions organized here will be instrumental in actively working towards these goals.

3. Flagship proposal 1980 of max 2000 words for 3a to 3d

3.a Ambitions

Our moonshot ambition is to shape the Consultation Room of the future by taking leadership in the transition towards digital healthcare. We work on this ambition on different levels (figure 1).

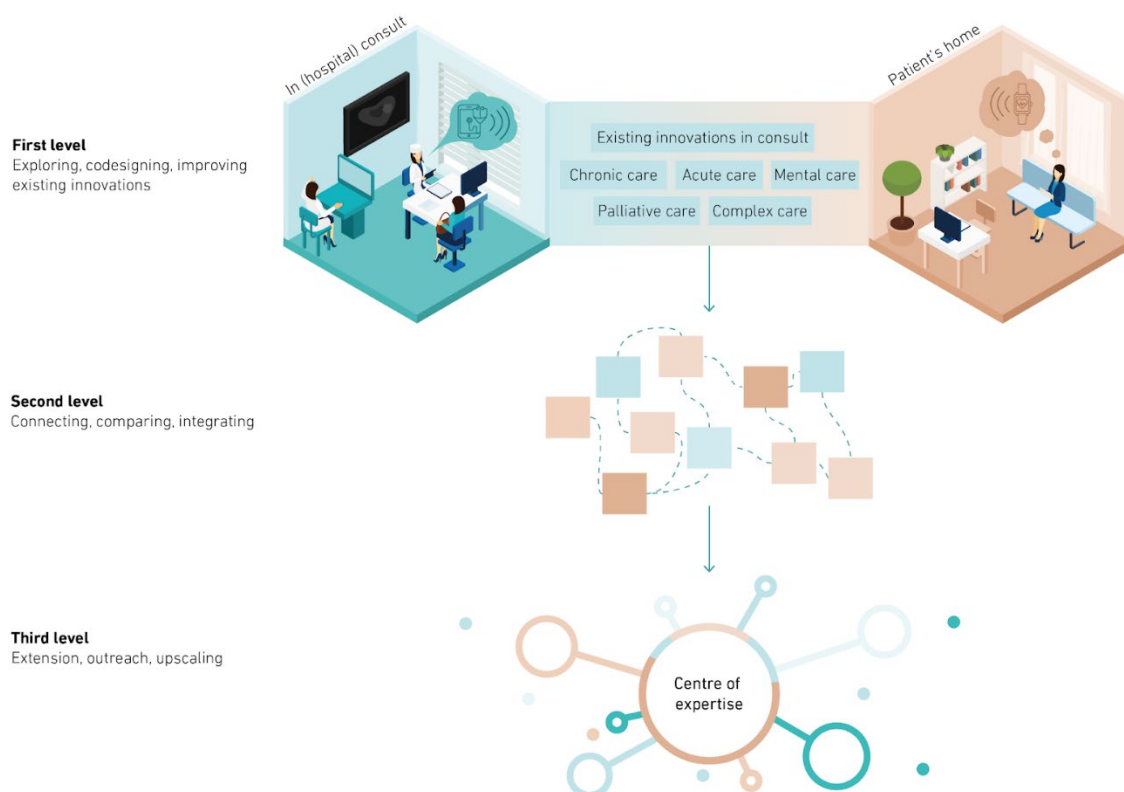


Figure 1 multi-level approach in the transition towards digital care

The **first level** relates to the **many high standard digital innovations that are already taking place** in the Erasmus MC within different departments. Consortium members in this Flagship are already collaborating for several years and many researchers are already involved in feasibility studies, pilot projects and validated innovations in digital healthcare (see 3b). In this Flagship, we build on these promising initiatives. Through facilitating co-design processes between clinicians, technology developers and patients in the five domains, we investigate how digital health technologies can be aligned with care processes and pathways, allowing existing initiatives to take the next step towards validation and clinical implementation.

The **second level** focuses on **connecting, comparing and integrating initiatives to identify more general lessons and best practices** and generate sustainable change. The innovations in level 1 relate to different domains but face similar challenges related to patient and professional expectations, IT- and data-infrastructures and structural forms of reimbursement. We study how healthcare processes need to be reconfigured for digital technologies to become part of 'regular' care, including changes related to reimbursement structures, IT-systems, and regulatory frameworks.

The **third level** focuses on **broader sustainability and outreach** of the Flagship. We expand our focus to include collaborations with other hospitals and care organizations in a national Centre of Expertise on digital health technologies in medicine. We will facilitate inter-organizational learning and identify the most promising technical, organizational, and regulatory solutions. We offer developers and users of digital health technologies the knowledge, methodological toolbox, and regulatory/organizational understanding to increase the acceptability and use of digital technologies.

3.b Aims and feasibility

Our moonshot ambition translates into three specific aims that together shape the (conditions for) continuity of care from hospital to home through digital technologies:

- 1) digital technologies for various conditions are integrated via co-design and validation in care pathways;
- 2) embedding of these technologies is addressed from an interdisciplinary perspective (taking into account challenges mentioned above);
- 3) impact of digital technologies is continuously monitored through formative evaluation accounting for different types of outcome measures and soft impacts.

Five clinical domains are integrated within the Flagship to have a broad variety of patient groups to design and use the technology for, providing major societal impact. For each domain, the Flagship is supported by matched funding of project leads.

Acute care	Safe @ home Pregnant women with a high risk of pregnancy complications need to visit the outpatient clinic up to three times a week. To lower the burden for these women, this program aims to monitor the women at home. The goal is to research the implementation and effects on care at a large scale. <i>Next steps to be co-designed are integration of the technology in the electronic patient file and suitable digital platforms and further evaluation from a value-based healthcare perspective (including safety).</i>
Chronic care	IPF online Patients with pulmonary fibrosis are mostly 65+, progressively ill, with a poor prognosis, decreasing mobility, on oxygen suppletion and using expensive drugs. We have shown that these patient like the home monitoring (including self performed wireless lung function measurements), are well able to do it, medication is adjusted more, and psychological wellbeing improves. <i>Next steps to be co-designed are how to structurally embed the system in existing clinical work and how to expand data collection (e.g., home measurements) and analysis (building algorithms to set alarms and predict exacerbations).</i>
Complex care	Digica Approximately 680,000 patients each year have preoperative anesthesia visits. The care for low-risk patients could be done at home, but current practice demands face-to-face informed consent. This project will develop a digital preoperative screening tool through which patients are informed, knowledge is tested, risk profiles are created, and informed consent is obtained. <i>Next steps to be co-designed are the organization of a 'smooth transition' of patients from the home environment to the hospital and organizing a data infrastructure that is aligned with existing data streams within the hospital.</i>
Mental care	EHealth Therapists started with digital psychotherapy programs, often used as blended-care with face-to-face or video consults. Development of usable interactive tools using emerging technologies such as sleep monitoring could help to overcome barriers in the patient-doctor interaction. <i>Next step to be co-designed is how to deal with these new sources of information in everyday clinical practice.</i>
Palliative care	Prediction modelling & Home-based care With individualized prognostic life expectancy models in the consultation room, head & neck cancer patients are empowered in making well considered end-of-life choices. Continuous remote care with easy to use and safe technology empowers patients to control their status and enables them to stay as long as possible with optimal quality of life in their preferred place: their home. <i>Next steps to be co-designed are preconditions for safe remote palliative care, user preferences of patients and professionals and development and testing of the home-based care path and the influence on optimal end-of-life care and patient empowerment.</i>

Figure 2 Examples of existing projects in digital health, and their next steps, that are already part of the consortium

Acute Care

The Department of Obstetrics and Gynaecology is working on Safe@home, research focusing on telemonitoring of pregnant women at risk for hypertensive pregnancy complications through a platform consisting of a wireless blood pressure monitor, app and integrated preeclampsia symptom checklist. Feasibility, patient experience and first evaluation studies showed many positive effects, including high compliance, no false-positive alerts, high satisfaction, and significantly fewer hospital admissions [7, 8].

Chronic Care

The Department of Pulmonary Medicine is working on various home monitoring projects for patients with pulmonary fibrosis. Studies show that programs with real-time wireless home spirometry for patients diagnosed with idiopathic pulmonary fibrosis (IPF) are reliable, produce positive patient outcomes, better adjusted medication and improved psychological well-being [9,22-24].

Complex Care

The Department of Anesthesiology is working on a digital preoperative screening tool to better differentiate between low-risk and high-risk patients to determine when preoperative anesthesia visits can be done at home. The department also established an AI-model that operates as triage system for identification of low and high-risk patients.

Mental Care

The Department of Psychiatry is working on automated analysis of speech and use of actigraphy to monitor sleep and movement for diagnosis and prediction of psychotic disorders in a large population-based cohort [25]. Psychiatric diagnostics will be strengthened by continuous behavior monitoring with digital wearables and collection of large amounts of data (e.g., for early detection of domestic violence).

Palliative Care

The Department of Otorhinolaryngology and Head and Neck Surgery is working on the structural implementation of remote palliative oncology care. Care at a distance allows patients in the palliative phase to receive optimal care in their trusted environment. An ePRO-structure is used to deliver remote symptom control via digital questionnaires and remote consultations with oncology nurses. EMC and TUD already developed a prototype app.

3.c Work plan

The project consists of three interrelated levels.

LEVEL 1: BUILDING FURTHER ON EXISTING INNOVATIONS			
Lead -- EMC Support -- TUD and EUR <i>To further develop and test digital health tools in medical practice through customized co-design in five domains</i>	WP 1	PhD 1	Acute care
		PhD 2	Chronic care
		PhD 3	Complex care
		PhD 4	Mental care
		PhD 5	Palliative care
LEVEL 2: CONNECTING AND INTEGRATING BEST PRACTICES TO CREATE SYSTEMIC CHANGE			
Lead -- EUR Support -- EMC and TUD <i>To address two of the most prominent overarching challenges to create systemic change</i>	WP 2	PhD 6	Sustainable reimbursement
	WP 3	PhD 7	Connected data- and IT- infrastructures
LEVEL 3: ESTABLISHING CENTRE OF EXPERTISE FOR SUSTAINABLE DIGITAL SOLUTIONS			
Lead -- EMC and EUR Support -- TUD <i>To expand our focus to include collaborations for broader sustainability</i>	WP 4	Post-doc	Formative evaluation
	WP 5		Community building
			Project management

LEVEL 1: BUILDING FURTHER ON EXISTING INNOVATIONS

WP 1 Tool development and testing in medical practice through customized co-design [PhD 1-5]

			Lead	Support	Resources	Partners
WP 1	PhD 1	Acute care	EMC	TUD + EUR	50 % own funding, 50 % Flagship	Partners and patients (representatives) from individual projects and collaborations
	PhD 2	Chronic care	EMC	TUD + EUR	50 % own funding, 50 % Flagship	
	PhD 3	Complex care	EMC	TUD + EUR	50 % own funding, 50 % Flagship	
	PhD 4	Mental care	EMC	TUD + EUR	50 % own funding, 50 % Flagship	
	PhD 5	Palliative care	EMC	TUD + EUR	50 % own funding, 50 % Flagship	
Aims					Output	
• Answer domain specific questions for digital health technologies • Connect existing initiatives to facilitate learning from other domains • Further develop and test digital health technologies with the use of co-design • Move beyond evaluations of individual projects with collaborations between PhDs • Align digital health technologies with existing pathways					• Digital health technologies embedded in specific context • Lessons learned from translating technologies to other domains • Co-designed (future) pathways • Challenges and lessons learned as input for WP2 and WP3 • Scientific output; peer-reviewed papers, 5 PhD theses, presentations or (scientific) meetings	

Aim:

The objective of WP 1 is to **further develop digital health tools and to test them in clinical practice through customized co-design** in the five domains. We use co-design methods with the tools developed and accepted by TU Delft for human-centered design and design for change [26] and by ESHPM from healthcare innovation [33-35] to target specific questions raised in the five clinical domains together with patients (see 3b).

Output:

WP1 consists of **5 PhDs** (1 co-funded PhD for each domain). This testifies to the Flagship alignment with existing medical questions and shows the support for the Flagship program in each EMC

department. Each PhD will focus on co-design methods to answer specific questions for (and tailor solutions to) each domain. We explicitly organize for cross-fertilization: every PhD is supervised by supervisors from at least 2 departments and supported by expertise from TUD and EUR.

LEVEL 2: CONNECTING AND INTEGRATING BEST PRACTICES TO CREATE SYSTEMIC CHANGE

The objective of WP 2 and 3 is to address two of the most prominent overarching challenges to create systemic change in the embedding of digital health technologies: **sustainable reimbursement structures** and **connected IT- and data-infrastructures**.

WP 2 Working towards sustainable reimbursement structures for digital technologies [PhD 6]

			Lead	Support	Resources	Partners
WP2	PhD 6	Sustainable reimbursement	EUR	TUD + EMC	100% Flagship	NZa, RIVM, health care insurer
Aims			Output			
- Support and participate in development of reimbursement procedures for the initiatives of WP1 - Application of time-driven activity-based costing in one of the digital health technologies' introductions of WP 1 - Evaluation of the introduction of alternative payment models in a field quasi experiment - Study of mechanisms in reimbursement that support the spread and upscaling of use of digital health technologies in a multiple longitudinal case study			- Scoping review of reimbursement procedures for digital health technologies in different health systems - Toolkit for applying time-driven activity-based costing on the introduction of a new technology - Webcast of generic lessons - Elearning of generic lessons for use in the Erasmus MC VBHC program and the NFU's Advanced Training Program VBHC - Scientific output; peer-reviewed papers, PhD thesis, presentations on (scientific) meetings			

Aim:

PhD 6 will focus on **novel reimbursement structures**. In recent years progress has been made with funding digital health technologies and developing reimbursement guidelines (e.g., through new DRG codes and payment titles). However, a lack of standardized, properly implemented national tariffs is still considered a persistent obstacle in widespread use of digital health technologies [36]. PhD 6 will participate in the co-design of digital health technologies from the perspective of payment models, focusing on embedding payment solutions for current and new technologies in all five domains, and on spread of working reimbursement structures for digital health technologies.

Output:

PhD 6 will deliver a scoping review of reimbursement procedures for digital health technologies in different health systems in a value-based healthcare context [31]. The PhD student will apply time-driven activity-based costing (TD-ABC) in one of the digital health technologies' introductions and evaluate the introduction of alternative payment models more suited for quality and collaboration, such as bundled payment and pay-for-performance. In a multiple longitudinal case study mechanisms in reimbursement supporting spread and upscaling of digital health technologies will be deciphered.

WP 3 Creating and maintaining connected data- and IT-infrastructures: Digitally Connected [PhD 7]

			Lead	Support	Resources	Partners
WP3	PhD 7	Connected data- and IT-infrastructures	EUR	EMC+TUD	100% Flagship	Digitaal verbonden
Aims					Output	
- Facilitate the provision of care at a distance by connecting all aspects of the care process - Large scale digital transformation of care pathways - Monitor the process of digitaal verbonden through ethnographic and qualitative research methods - To give continuous feedback on the process of digitaal verbonden					- A roadmap on how data and IT infrastructures can be organized to optimally support the sustainable embedding of digital health technologies - An overview of recommendations to maximize intervention effectiveness and implementation success which can be used in WP 1 - Infographics and short knowledge clips with key lessons learned - Scientific output; peer-reviewed papers, PhD thesis, presentations on (scientific) meetings	

Aim:

PhD 7 will focus on **novel data and IT-infrastructures**. We closely collaborate with the Erasmus MC-wide strategic program Digitally Connected, which provides in-kind contribution to this WP. This program aims to facilitate provision of care at a distance by connecting all aspects of the care process (preparation, intake, communication, monitoring, treatment). This program will be ongoing until at least 2027 and will focus on **large scale digital transformation of up to 350 care pathways**. PhD 7 will closely monitor key aspects of this process, including data ethics, through ethnographic and qualitative research methods in close alignment with the post-doc focusing on formative evaluation. The action-oriented approach allows for regular feedback moments that will help improve the program.

Output:

PhD 7 will provide a roadmap on supportive data- and IT-infrastructures to facilitate sustainable embedding of digital health technologies and create an overview of recommendations to maximize intervention effectiveness and implementation success. A deliverable is also a PhD thesis with various academic publications, focused on expectations of professionals and patients regarding data and IT-support, and data ethics. Deliverables are infographics and short knowledge clips with key lessons learned.

LEVEL 3: ESTABLISHING CENTRE OF EXPERTISE FOR SUSTAINABLE DIGITAL SOLUTIONS

WP 4 Continuous formative evaluation [Post-Doc]

			Lead	Support	Resources	Partners
WP4	Post-doc	Continuous formative evaluation	EUR	EMC+TUD	100% Flagship	Digitaal verbonden, reflective board
Aims					Output	
- Providing a formative evaluation that accompanies the initiatives in this Flagship - Identifying broader lessons and insights of WP 1, 2 and 3 - Steer the projects of WP 1 dynamically, depending on the insights gained during the process - Monitor development of tradeoffs between different goals and values of digital health technologies (e.g. efficiency, patient satisfaction, workflow)					- Method that can take into account the dynamic nature of technological change - Academic publications, focused on the emerging insights that cut across the different pilot projects - Infographics and short knowledge clips with key lessons learned - Interdisciplinary evaluation tool for technologies in healthcare which can account for different types of outcome measures and soft impact - Scientific output; peer-reviewed papers, PhD thesis, presentations on (scientific) meetings	

Aim:

Whereas PhD 1-5 focus on the effects of individual projects on health outcomes and (cost-) effectiveness of interventions and tools, this post-doc conducts **a formative evaluation of initiatives**

in this Flagship, identifying and sharing broader lessons and insights. Crucially, this formative evaluation does not aim to evaluate the projects after they are finished but focuses on 'situated intervention' [27], allowing more dynamic feedback of insights gained. The post-doc will focus on less tangible changes and tradeoffs triggered by the shift towards digital technologies [20, 30].

Output:

This post-doc focuses on developing a method that considers the dynamic nature of technological change. The output will serve as regular feedback of insights into the program. This post-doc will deliver various academic publications, focused on emerging insights transcending the different pilot projects. Deliverables comprise infographics and short knowledge clips with key lessons.

WP 5 Project management and community building

			Resources	Partners
WP5	Postdoc	Project management and community building	EMC+TUD+EUR	100% Flagship
Aims			Output	
- Community building - Smooth and efficient organization - Adherence to deadlines and deliverables - Identify risks and opportunities in an early stage - Identify and proactively act upon funding opportunities - Enhance external visibility of the Flagship			- Reflective board of legal experts, ethicists, privacy officers, as well as representatives of regulatory bodies - Website as platform for further communication during the Flagship. We will post regular updates, newsletters, links to relevant symposia and conferences, and other relevant meetings - Convergence hubs where PhDs, students and postdocs can work on their own projects while also creating opportunities to exchange ideas and develop novel combinations - Bimonthly 'Inspiration Meetings' - Presentations on (scientific) meetings about the consultation room 2030	

WP5 focuses on project management to enable smooth and efficient organization and collaboration, adherence to deadlines and deliverables, and identify risks and opportunities in an early stage. To ensure efficient operational governance of the Flagship, the post-doc will coordinate monthly meetings for PhDs and post-docs. A key aim of this work package is also **community building**. This is done through the four formats identified elaborated in 4d: Reflective Board, website, Convergence Hubs, and Inspiration Meetings, which include students from all institutions.

3.d Alignment of the program with relevant national and international research agendas

There is increasing pressure on the financial, personnel and societal sustainability of the Dutch healthcare system [1]. Healthcare must develop into an integrated, proactive, value-based, personalized and home-centered approach with more focus on prevention of disease and positive lifelong health. Digital health technologies play a major role in this transition [29]. Continuous self-monitoring by patients allows care professionals to provide care at the right time (preferably early), the right place (where possible at home) and in the right way (including prevention). The Knowledge Platform 'Right care in the Right Place' [41] argues that the main problems of this transition lie in issues of collaboration and reimbursement; this flagship is expected to find ZonMw funding connected to this agenda.

To harvest the potential of digital health technologies is also a broad policy aim in the Netherlands as in Europe. A core recommendation of the Social Economic Council (SER) regarding sustainability of the future healthcare system focuses on taking necessary steps to enhance use of digital

innovations by patients and professionals [2]. A main aim in the current knowledge agendas of Health Holland and MedTech NL is to stimulate the use of digital technologies to organize care and support for people at home, preventing unnecessary hospital visits and enabling monitoring at a distance. A recent letter to parliament contains a plea for using more technology in providing aftercare at home [42]. On a European level we witness a call for active strategies to support the current momentum around the use of digital health technologies [6].

4. Consortium 1990 of max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

The consortium consists of 31 members; due to the word limit we only list some key members here; the full list of consortium members can be found at the frontpage.

Flagship leads

Joke Hendriks is Professor of Surgery and chair of the department of Surgery. She wants to shape the Consultation Room of the future by taking leadership in the transition towards digital health care. For the past two years she was the coordinator of this project and brought together many researchers in this field.

Richard Goossens is Professor of Physical Ergonomics at the Faculty of Industrial Design Engineering. The emphasis of his research is on ergonomics / human factors in relation to product innovation in order to establish safe, comfortable and error-free task fulfillment in complex multi-user situations in a medical setting. The goal is to generate design requirements, methods and product (services). The requirements should anticipate the implementation of these products on a large scale and daily use.

Kees Ahaus is professor of Health Service Management & Organisation (HSMO) at ESHPM and main lead on behalf of EUR for this flagship. His research interest is value-based health care (including reimbursement of care). One of the research topics of his department HSMO is eHealth (such as remote monitoring), big data/AI and adoption of new technology. He is chairman of the knowledge platform Right Care in the Right Place instigated by the Ministry of Health.

Erasmus University Medical Centre

Eveline Corten is a reconstructive plastic surgeon. Mid-2019, she started her research on Extended Reality (XR) for patient experience, preoperative planning, and medical education. In 2021 she received two grants (€ 810.000) for her projects. She has a network of people with various backgrounds and expertise in social sciences, industrial design, and computational imaging.

Arie Franx is full Professor of Obstetrics. He supervised over 60 PhD students and co-authored over 300 scientific papers and two handbooks. He initiated and supervised a longitudinal research line on telemonitoring of high-risk pregnancies. He also works on Value Based Health Care with the Netherlands Federation of University Medical Centres, the Ministry of Health and the Netherlands Healthcare Institute and the International Consortium of Health Outcome Measurement (www.ichom.org).

Witte Hoogendijk a psychiatrist and professor of psychiatry. He chairs the Epidemiological and Social Research institute, a collaboration of mental institutions and the municipality of Rotterdam that form an infrastructure for data science in mental health care. One of the research topics of his department is digitalization of diagnostic procedures and psychiatric treatments.

Jan-Wiebe Korstanje is a biomedical engineer, anesthesiologist and assistant professor with a special focus on big-data and the transformation of a physical outpatient clinical towards a home-centered clinic using AI and smart monitoring in the perioperative setting starting from home but not ending when returning to their home.

Marinella Offerman is Assistant Professor, psychologist and sector head Value Based Healthcare and palliative and psychosocial care for Head & Neck cancer. She leads the research line for patient centered care and individualized prognostic counseling. Recent major funding for her research line were: €574.007,80 KWF (2021-2025) and €924.983,00 Zorginstituut (2018-2021).

Jan-Jaap Visser is trained as a radiologist and involved in quality and ICT-developments within radiology both nationally and internationally. He is also Chief Medical Information Officer (CMIO) at EMC. With his clinical and ICT experience, he will fulfill the important bridging function between ICT and care. He is currently involved in the Citrien eHealth program for scaling up eHealth initiatives. He is member of the steering committee on the institutional program to digitize patient care and facilitate home monitoring.

Marlies Wijsenbeek is Associate Professor, pulmonologist and head of the Academic Centre of Excellence for Interstitial Lung Diseases (ILD) and Sarcoidosis and chair of the ILD group of the European Respiratory Society. Together with patients she developed a home monitoring system integrating also home-based lung function measurements. She leads several (inter)national home-monitoring research projects.

Delft University of Technology

Maike Kleinsmann is Professor in Design for Digital Transformation at the Faculty of Industrial Design Engineering. Her expertise is on collaborative design. She founded the Cardiolab fieldlab together with De Hartstichting and Philips Healthcare. She is also the 'kwartiermaker' of the Cardiology Tech line of the Delft Health Initiative. and board member of the National e-health Living Lab (NeLL). She also works on the Convergence Alliance 'Healthy Start' with the Erasmus MC and Erasmus University.

John Clarkson is Professor of Healthcare Systems at Delft University of Technology, Professor of Engineering Design at the University of Cambridge, Director of the Cambridge Engineering Design Centre and Co-Director of Cambridge Public Health. His research interests are in the general area of engineering design, particularly the development of design methodologies to support healthcare design.

Erasmus University Rotterdam

Roland Bal is professor of Healthcare governance at ESHPM. He has done many studies on ICT use in integrated care, following a formative evaluation approach. He works closely with the Healthcare Inspectorate and the Dutch healthcare Institute on the regulation of quality of care, and of technology.

Rik Wehrens is assistant professor at ESHPM, with expertise in science & technology studies and ethics. He utilizes ethnographic and qualitative research methods in formative evaluation and action research. Current research focuses on implications of digital and data-driven health. He was part of the recently finished EU-funded project Big Medilytics (<https://www.bigmedilytics.eu>).

Petra Porte is an assistant professor at ESHPM in the field of technological innovations in healthcare, with an expertise in clinical technologies and clinical informatics. Current research focuses on data driven work in the hospital and evaluation of home-monitoring tools and other technological innovations.

Evert F. Stamhuis (LLM, PhD) holds a chair for Law and Innovation at Erasmus School of Law and is Senior Fellow of the Jean Monnet Centre of Excellence on Digital Governance. He researches the interaction between law, governance and new digital technologies. He is co-applicant of the recently awarded AI-MAPS ELSA Lab (€2.1 mln) and is invited expert in many consortia.

Mauren Rutten is professor of economic evaluation of innovations for health and scientific director of the Institute for Medical Technology Assessment (IMTA). Her expertise includes economic evaluation studies and outcomes research. She has a special interest in the cost-effectiveness, and the financing and payment of complex multi-faceted interventions that involve system-level changes like hospital at home and remote monitoring programs. She is coordinator of various Horizon2020 projects.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The overall moonshot aim of the Flagship is taking leadership in the transition towards digital healthcare through a multi-level approach. This requires a transdisciplinary approach that includes medical, technical, design, sociological, organizational, health economic, governance, and regulatory experts, which are all involved. We work on this ambition on three levels (for details see 3a and 3b).

The first level relates to the many high standard digital innovations that are already taking place. We involve experts from five domains. Prof. Franx, dr. Korstanje, prof. Hoogendijk, prof. Offerman and dr. Wijsenbeek are all established experts in their respective medical domains. They are leads of the clinical projects that form the basis of WP1. The in-cash contribution of 50% testifies to how the program aligns with existing medical questions and needs, and shows the support for the Flagship program in each department of the EMC. Prof. Goossens, Prof. Kleinsmann, dr. Tabeau, dr. Jung and dr. Giunti add valuable expertise in design for impact and (co-design) methods for sustainable implementation. Prof. Rutten is expert in cost-effectiveness studies.

The second level focuses on connecting, comparing and integrating initiatives to identify more general lessons and best practices and generate sustainable organizational change. Prof. Bal and dr. Wallenburg are experts in the field of action-research, and in collaborating with healthcare organizations and public regulatory agencies. Prof. Ahaus is expert in value-based health care and reimbursement of care. Dr. Wehrens is expert in the adoption of complicated technologies in healthcare practice and action-oriented research and dr. Porte is expert in clinical technologies and clinical informatics.

The third level focuses on broader sustainability and outreach of the Flagship. All members have a proven track record on outreach, valorisation and acquisition, as can be testified through the amount of external grants, connections to industry, and membership in national and international expert bodies the consortium members bring.

4.c Involvement of partners

In the different levels of the flagship, partners are involved in different ways. The existing projects of level 1 have **individual partnerships for projects**. These partnerships are, for example, with commercial companies, patient associations and other institutes. Researchers from this Flagship are experienced within their field of research and will discuss and connect with stakeholders to further develop the research within the Flagship. Individual projects are supported or funded by their

individual partnerships, which we want to build upon for future developments within the Flagship. At this moment we collaborate with, among others, SyncVR Medical, Luscii, NovaCair, SkinVision, Curavista, Attendi, patiëntenvereniging Hoofd-Hals, the Dutch pulmonary fibrosis association, and the European Pulmonary Fibrosis Federation. In level 2 we have involved ***Digitaal verbonden*** and ***representatives from regulatory bodies***. To address systemic issues a broader perspective is necessary, and several stakeholders showed intent or commitment to connect to our Flagship. In level 3 we aim for a broader sustainability and outreach with a ***reflective board*** and ***Digitaal verbonden***. For this we have involved various external partners, such as the National Health Care Institute, Dutch Healthcare Authority, insurance companies, non-academic hospitals, MedTech and social enterprises. External partners take place in the reflective board (see 4d) or are connected to the project of Digitaal verbonden which aims to connect with other health care providers for information exchange.

4.d Governance, organization and collaboration strategy of the consortium

The Flagship leads of EMC, TUD and EUR together with core staff members of each institute will constitute the ***Management Board [MB]*** responsible for overall development of the Flagship. The Management Board will meet 6 times per year to discuss progress and identify opportunities. They will also coordinate interim progress reports for the convergence board. A major responsibility will be to identify and proactively act upon funding opportunities, and to enhance external visibility of the Flagship.

External partners are engaged through the ***Reflective Board*** (see 2d) and our research projects. The Reflective Board consists of governance and regulatory actors (Prof. Leistikow, Prof. Delnoij), health insurers (Van Hoof, Struis), members focused on data protection and ethics (Prof. Stamhuis), and representatives from patient and professional associations, who meet two times per year to discuss developments and progress of the Flagship and provide feedback and recommendations to the Management Board and to specific projects.

A key aspect of the synergy we seek to achieve relates to the creation of spaces where cross-fertilization can occur and exciting new ideas can be formed. To facilitate this, we create ***'Convergence Hubs'*** where PhDs, students and postdocs can work on their own projects while also creating opportunities to exchange ideas and develop novel combinations. PhDs and postdocs in particular are expected to work at least one day per week from these Hubs. Providing space for new ideas and projects also requires a partial move away from the 'projectified' logic [21] which requires detailed planning and milestones. While we have a clear work plan with deliverables and milestones (see 3b and 3c), we deliberately seek to create some leeway for the PhDs and postdocs as to allow them to work on new ideas, methods and collaborative projects. The ***creation of multidisciplinary supervision teams*** also allows for further synergy in the Flagship.

A ***website*** is already under construction, with information about involved researchers, areas of expertise, previous related projects and new projects and initiatives (www.spreekkamer2030.nl). This website forms the platform for further communication during the Flagship. We will post regular updates, newsletters, links to relevant symposia and conferences. We will also organize bimonthly Inspiration Meetings: informal, easily accessible meetings around specific topics that are of relevance to the broader digital health community.

Annexes

I Budget table (including the external funding table)

II Letters of intent and letters of commitment

Letter of commitment:

- Digitaal Verbonden

Letters of intent for collaboration:

- SyncVR medical
- Patiëntenvereniging hoofd-hals
- Patiëntenvereniging longfibrose
- European Pulmonary Fibrosis Federation (EU-IPFF)
- ESPRi
- ESHPM

Letters of intent for reflective board:

- Geert van Hoof
- Jeroen Struijs
- Peter Dohmen

III Letter of support from department head of the main lead

IV References